

Part A - Basic grep and Filtering

- Display all lines containing the word failed.

grep "failed" security_log.txt

```
(praneetha@Praneetha)-[~/Desktop]
$ grep "failed" security_log.txt
```

- Display all lines containing failed, ignoring case.

grep -i "failed" security_log.txt

```
(praneetha@Praneetha)-[~/Desktop]
$ grep -i "failed" security_log.txt
[2025-08-18 08:12:03] Failed login for user root from 203.0.113.45 port 4444
[2025-08-18 08:12:10] Failed login for user root from 203.0.113.45 port 4444
[2025-08-18 08:12:17] Failed login for user root from 203.0.113.45 port 4444
[2025-08-18 08:29:58] Failed login for user guest from 10.1.2.3 port 22
[2025-08-18 08:30:17] Failed login for user guest from 10.1.2.3 port 22
[2025-08-18 08:30:44] Failed login for user admin from 172.16.5.10 port 2222
[2025-08-18 08:31:03] Failed login for user admin from 172.31.200.9 port 2222
[2025-08-18 08:42:01] Failed login for user root from 198.51.100.77 port 2222
[2025-08-18 08:42:19] Failed login for user root from 198.51.100.77 port 2222
[2025-08-18 08:42:38] Failed login for user admin from 198.51.100.77 port 2222
```

- Display all lines that contain either Failed or Denied.

grep -E "Failed|Denied" security_log.txt

```
(praneetha@Praneetha)-[~/Desktop]
$ grep -E "Failed|Denied" security_log.txt
[2025-08-18 08:12:03] Failed login for user root from 203.0.113.45 port 4444
[2025-08-18 08:12:10] Failed login for user root from 203.0.113.45 port 4444
[2025-08-18 08:12:17] Failed login for user root from 203.0.113.45 port 4444
[2025-08-18 08:29:58] Failed login for user guest from 10.1.2.3 port 22
[2025-08-18 08:30:17] Failed login for user guest from 10.1.2.3 port 22
[2025-08-18 08:30:44] Failed login for user admin from 172.16.5.10 port 2222
[2025-08-18 08:31:03] Failed login for user admin from 172.31.200.9 port 2222
[2025-08-18 08:42:01] Failed login for user root from 198.51.100.77 port 2222
[2025-08-18 08:42:19] Failed login for user root from 198.51.100.77 port 2222
[2025-08-18 08:42:38] Failed login for user admin from 198.51.100.77 port 2222
```

- Display all lines that do not contain the word INFO.

grep -v "INFO" security_log.txt

```
(praneetha@Praneetha)-[~/Desktop]
$ grep -v "INFO" security_log.txt
[2025-08-18 08:12:03] Failed login for user root from 203.0.113.45 port 4444
[2025-08-18 08:12:10] Failed login for user root from 203.0.113.45 port 4444
[2025-08-18 08:12:17] Failed login for user root from 203.0.113.45 port 4444
[2025-08-18 08:13:02] WARNING Multiple authentication failures from 203.0.113.45
[2025-08-18 08:15:44] ERROR Connection denied from 198.51.100.77 port 9001
[2025-08-18 08:16:05] Suspicious connection from 198.51.100.77 to evilc2.ddns.net:9001
[2025-08-18 08:16:19] Suspicious outbound request https://evilc2.ddns.net/update.php
[2025-08-18 08:18:07] WARNING Request to http://malware-test.example.org/download.php from 1
[2025-08-18 08:18:20] Suspicious file download evil.exe from http://malware-test.example.org
[2025-08-18 08:18:32] ERROR file evil.exe hash MD5=44d88612fea8a8f36de82e1278abb02f
[2025-08-18 08:19:45] Suspicious file invoice_2025.pdf downloaded from https://phishing-exam
[2025-08-18 08:20:10] WARNING Email received from attacker123@phishing-example.com
[2025-08-18 08:20:51] Suspicious email phishing.alert@evilmail.ru contained malicious attach
[2025-08-18 08:22:03] ERROR Exploit attempt CVE-2025-12345 from 203.0.113.99
[2025-08-18 08:22:21] WARNING Vulnerability CVE-2024-3094 detected on server
```

- Display all lines containing the exact word root.

grep -w "root" security_log.txt

```
(praneetha@Praneetha)-[~/Desktop]
$ grep -w "root" security_log.txt
[2025-08-18 08:12:03] Failed login for user root from 203.0.113.45 port 4444
[2025-08-18 08:12:10] Failed login for user root from 203.0.113.45 port 4444
[2025-08-18 08:12:17] Failed login for user root from 203.0.113.45 port 4444
[2025-08-18 08:27:24] Suspicious crontab created for user root
[2025-08-18 08:42:01] Failed login for user root from 198.51.100.77 port 2222
[2025-08-18 08:42:19] Failed login for user root from 198.51.100.77 port 2222
```

- Display all occurrences of the word attack along with their line numbers.

grep -n "attack" security_log.txt

```
(praneetha@Praneetha)-[~/Desktop]
$ grep -n "attack" security_log.txt
17:[2025-08-18 08:20:10] WARNING Email received from attacker123@phishing-example.com
```

- Count the number of lines containing ERROR.

grep -c "ERROR" security_log.txt

```
(praneetha@Praneetha)-[~/Desktop]
$ grep -c "ERROR" security_log.txt
10
```

- Display only the filenames of files whose contents contain the word malware, assuming multiple log files are available.

grep -l "malware" *.txt

```
(praneetha@Praneetha)-[~/Desktop]
$ grep -l "malware" *.txt
security_log.txt
```

- Display the first five lines containing Suspicious.

grep "Suspicious" security_log.txt | head -5

```
(praneetha@Praneetha)-[~/Desktop]
$ grep "Suspicious" security_log.txt | head -5
[2025-08-18 08:16:05] Suspicious connection from 198.51.100.77 to evilc2.ddns.net:9001
[2025-08-18 08:16:19] Suspicious outbound request https://evilc2.ddns.net/update.php
[2025-08-18 08:18:20] Suspicious file download evil.exe from http://malware-test.example.org/evil.exe
[2025-08-18 08:19:45] Suspicious file invoice_2025.pdf downloaded from https://phishing-example.com/invoice_2025.pdf
[2025-08-18 08:20:51] Suspicious email phishing.alert@evilmail.ru contained malicious attachment
```

- Display the last five failed-login entries.

grep -i "Failed login" security_log.txt | tail -5

```
(praneetha@Praneetha)-[~/Desktop]
$ grep -i "Failed login" security_log.txt | tail -5
[2025-08-18 08:30:44] Failed login for user admin from 172.16.5.10 port 2222
[2025-08-18 08:31:03] Failed login for user admin from 172.31.200.9 port 2222
[2025-08-18 08:42:01] Failed login for user root from 198.51.100.77 port 2222
[2025-08-18 08:42:19] Failed login for user root from 198.51.100.77 port 2222
[2025-08-18 08:42:38] Failed login for user admin from 198.51.100.77 port 2222
```

Part B - Regular Expression Extraction

- Extract all IPv4 addresses from the log.

grep -Eo '([0-9]{1,3}\.){3}[0-9]{1,3}' security_log.txt

```
(praneetha@Praneetha)~[/Desktop]
$ grep -Eo '([0-9]{1,3}\.){3}[0-9]{1,3}' security_log.txt
192.168.10.5
192.168.10.21
203.0.113.45
203.0.113.45
203.0.113.45
203.0.113.45
10.10.5.23
198.51.100.77
198.51.100.77
192.168.10.21
10.20.30.40
203.0.113.99
10.1.2.3
10.1.2.3
10.1.2.3
10.1.2.3
172.16.5.10
172.31.200.9
192.168.1.50
```

- Extract all IP addresses belonging to the 10.x.x.x private network.

grep -Eo '\b10\.([0-9]{1,3}\.){2}[0-9]{1,3}\b' security_log.txt

```
(praneetha@Praneetha)~[/Desktop]
$ grep -Eo '\b10\.([0-9]{1,3}\.){2}[0-9]{1,3}\b' security_log.txt
10.10.5.23
10.20.30.40
10.1.2.3
10.1.2.3
10.1.2.3
10.1.2.3
```

- Extract all addresses belonging to the 192.168.x.x network.

grep -Eo '\b192\.168\.([0-9]{1,3}\.){0}[0-9]{1,3}\b' security_log.txt

```
(praneetha@Praneetha)~[/Desktop]
$ grep -Eo '\b192\.168\.([0-9]{1,3}\.){0}[0-9]{1,3}\b' security_log.txt
192.168.10.5
192.168.10.21
192.168.10.21
192.168.1.50
192.168.10.5
192.168.10.5
192.168.10.21
192.168.10.21
```

- Extract all email addresses.

grep -Eo '[[[:alnum:]]._%+~]+@[[[:alnum:]].-]+\.[[:alpha:]]+' security_log.txt

```
(praneetha@Praneetha) - [~/Desktop]
$ grep -Eo '[[[:alnum:]]._%+~]+@[[[:alnum:]].-]+\.[[:alpha:]]+' security_log.txt
attacker123@phishing-example.com
security.team@example.com
phishing.alert@evilmail.ru
student2025@university.in
```

- Extract only email addresses whose domain ends in .com.

grep -Eo '[[[:alnum:]]._%+~]+@[[[:alnum:]].-]+\.[[:alpha:]]+\.com\b' security_log.txt

```
(praneetha@Praneetha) - [~/Desktop]
$ grep -Eo '[[[:alnum:]]._%+~]+@[[[:alnum:]].-]+\.[[:alpha:]]+\.com\b' security_log.txt
attacker123@phishing-example.com
security.team@example.com
```

- Extract email addresses whose username contains at least one digit.

grep -Eo '[[[:alnum:]]._%+~]*[0-9]([[[:alnum:]]._%+~]*@[[[:alnum:]].-]+\.[[:alpha:]]+)' security_log.txt

```
(praneetha@Praneetha) - [~/Desktop]
$ grep -Eo '[[[:alnum:]]._%+~]*[0-9]([[[:alnum:]]._%+~]*@[[[:alnum:]].-]+\.[[:alpha:]]+)' security_log.txt
attacker123@phishing-example.com
student2025@university.in
```

- Extract all URLs beginning with http:// or https://.

grep -Eo 'https?://[^\[:space:]]+' security_log.txt

```
(praneetha@Praneetha) - [~/Desktop]
$ grep -Eo 'https?://[^\[:space:]]+' security_log.txt
https://evilc2.ddns.net/update.php
http://malware-test.example.org/download.php
http://malware-test.example.org/evil.exe
https://phishing-example.com/invoice_2025.pdf
https://target.example.com/login.php
https://www.example.com/index.html
https://shop.example.com/admin.php
http://legacy.example.net/login.php
https://portal.university.in/login.php
https://updates.bad-domain.ru/update.php
https://updates.bad-domain.ru/update.php
http://evil.example.net/dropper.exe
https://files.example.com/upload.php
https://command.bad-domain.in/backdoor.exe
```

- Extract only HTTPS URLs.

grep -Eo 'https://[^\[:space:]]+' security_log.txt

```
(praneetha@Praneetha) - [~/Desktop]
$ grep -Eo 'https://[^\[:space:]]+' security_log.txt
https://evilc2.ddns.net/update.php
https://phishing-example.com/invoice_2025.pdf
https://target.example.com/login.php
https://www.example.com/index.html
https://shop.example.com/admin.php
https://portal.university.in/login.php
https://updates.bad-domain.ru/update.php
https://updates.bad-domain.ru/update.php
https://files.example.com/upload.php
https://command.bad-domain.in/backdoor.exe
```

- Extract URLs ending with .php.

grep -Eo 'https?:/[^\[:space:]]+\.\php\b' security_log.txt

```
(praneetha@Praneetha) - [~/Desktop]
$ grep -Eo 'https?:/[^\[:space:]]+\.\php\b' security_log.txt
https://evilc2.ddns.net/update.php
http://malware-test.example.org/download.php
https://target.example.com/login.php
https://shop.example.com/admin.php
http://legacy.example.net/login.php
https://portal.university.in/login.php
https://updates.bad-domain.ru/update.php
https://updates.bad-domain.ru/update.php
https://files.example.com/upload.php
```

- Extract all CVE identifiers in the form:CVE-YYYY-NNNN

grep -Eo 'CVE-[0-9]{4}-[0-9]{4,}' security_log.txt

```
(praneetha@Praneetha) - [~/Desktop]
$ grep -Eo 'CVE-[0-9]{4}-[0-9]{4,}' security_log.txt
CVE-2025-12345
CVE-2024-3094
CVE-2025-987654
CVE-2023-44487
CVE-2025-54321
CVE-2025-54321
```

- Extract only CVEs reported for the year 2025.

grep -Eo 'CVE-2025-[0-9]{4,}' security_log.txt

```
(praneetha@Praneetha) - [~/Desktop]
$ grep -Eo 'CVE-2025-[0-9]{4,}' security_log.txt
CVE-2025-12345
CVE-2025-987654
CVE-2025-54321
CVE-2025-54321
```

- Extract all 32-character hexadecimal MD5 hashes.

grep -Eo '\b[[:xdigit:]]{32}\b' security_log.txt

```
(praneetha@Praneetha) - [~/Desktop]
$ grep -Eo '\b[[:xdigit:]]{32}\b' security_log.txt
44d88612fea8a8f36de82e1278abb02f
e2fc714c4727ee9395f324cd2e7f331f
5d41402abc4b2a76b9719d911017c592
```

- Extract all 40-character hexadecimal SHA-1 hashes.

grep -Eo '\b[[:xdigit:]]{40}\b' security_log.txt

```
(praneetha@Praneetha) - [~/Desktop]
$ grep -Eo '\b[[:xdigit:]]{40}\b' security_log.txt
3395856ce81f2b7382dee72602f798b642f14140
```

- Extract all 64-character hexadecimal SHA-256 hashes.

grep -Eo '\b[[:xdigit:]]{64}\b' security_log.txt

```
(praneetha@Praneetha) - [~/Desktop]
$ grep -Eo '\b[[:xdigit:]]{64}\b' security_log.txt
275a021bbfb6489e54d471899f7db9d1663fc695ec2fe2a2c4538aabf651fd0f
3a7bd3e2360a3d80a9d0b7b93d7f90d7c0b8c9a3e9f1d2c4b5a6978877665544
```

- Extract all timestamps having the format:

YYYY-MM-DD HH:MM:SS

grep -Eo '[0-9]{4}-[0-9]{2}-[0-9]{2} [0-9]{2}:[0-9]{2}:[0-9]{2}' security_log.txt

```
(praneetha@Praneetha)-[~/Desktop]
$ grep -Eo '[0-9]{4}-[0-9]{2}-[0-9]{2} [0-9]{2}:[0-9]{2}:[0-9]{2}' security_log.txt
2025-08-18 08:10:01
2025-08-18 08:11:14
2025-08-18 08:12:03
2025-08-18 08:12:10
2025-08-18 08:12:17
2025-08-18 08:13:02
2025-08-18 08:14:33
2025-08-18 08:15:44
2025-08-18 08:16:05
2025-08-18 08:16:19
2025-08-18 08:17:31
2025-08-18 08:18:07
2025-08-18 08:18:20
```

Part C - Anchors and Regex Patterns

- Display all lines beginning with a timestamp.

grep -E '^[[0-9]{4}-[0-9]{2}-[0-9]{2} [0-9]{2}:[0-9]{2}:[0-9]{2}\]' security_log.txt

```
(praneetha@Praneetha)-[~/Desktop]
$ grep -E '^[[0-9]{4}-[0-9]{2}-[0-9]{2} [0-9]{2}:[0-9]{2}:[0-9]{2}\]' security_log.txt
[2025-08-18 08:10:01] INFO Server startup completed on 192.168.10.5 port 8080
[2025-08-18 08:11:14] INFO User admin logged in from 192.168.10.21
[2025-08-18 08:12:03] Failed login for user root from 203.0.113.45 port 4444
[2025-08-18 08:12:10] Failed login for user root from 203.0.113.45 port 4444
[2025-08-18 08:12:17] Failed login for user root from 203.0.113.45 port 4444
[2025-08-18 08:13:02] WARNING Multiple authentication failures from 203.0.113.45
[2025-08-18 08:14:33] INFO Successful login for user analyst from 10.10.5.23
[2025-08-18 08:15:44] ERROR Connection denied from 198.51.100.77 port 9001
[2025-08-18 08:16:05] Suspicious connection from 198.51.100.77 to evilc2.ddns.net:9001
[2025-08-18 08:16:19] Suspicious outbound request https://evilc2.ddns.net/update.php
[2025-08-18 08:17:31] INFO DNS query for portal.example.com from 192.168.10.21
[2025-08-18 08:18:07] WARNING Request to http://malware-test.example.org/download.php from 1
[2025-08-18 08:18:20] Suspicious file download evil.exe from http://malware-test.example.org
[2025-08-18 08:18:33] ERROR File evil.exe hash MD5:44d886125ea8a8f26de82a1278abb82f
```

- Display all lines beginning with the word Suspicious.

grep '^Suspicious' security_log.txt

```
(praneetha@Praneetha)-[~/Desktop]
$ grep '^Suspicious' security_log.txt
Suspicious activity detected from 203.0.113.45
Suspicious malware file evil.exe
```

- Display lines ending with a numeric value.

grep -E '[0-9]+\$' security_log.txt

```
(praneetha@Praneetha)-[~/Desktop]
$ grep -E '[0-9]+$' security_log.txt
[2025-08-18 08:10:01] INFO Server startup completed on 192.168.10.5 port 8080
[2025-08-18 08:11:14] INFO User admin logged in from 192.168.10.21
[2025-08-18 08:12:03] Failed login for user root from 203.0.113.45 port 4444
[2025-08-18 08:12:10] Failed login for user root from 203.0.113.45 port 4444
[2025-08-18 08:12:17] Failed login for user root from 203.0.113.45 port 4444
[2025-08-18 08:13:02] WARNING Multiple authentication failures from 203.0.113.45
[2025-08-18 08:14:33] INFO Successful login for user analyst from 10.10.5.23
[2025-08-18 08:15:44] ERROR Connection denied from 198.51.100.77 port 9001
[2025-08-18 08:16:05] Suspicious connection from 198.51.100.77 to evilc2.ddns.net:9001
[2025-08-18 08:17:31] INFO DNS query for portal.example.com from 192.168.10.21
```

- Display lines beginning with ERROR or WARNING.

grep -E '^(ERROR|WARNING)' security_log.txt

```
(praneetha@Praneetha)~[~/Desktop]
$ grep -E '^(ERROR|WARNING)' security_log.txt
ERROR service terminated with code 137
WARNING connection retry 3
```

- Display lines whose first character is a digit.

grep -E '^[0-9]' security_log.txt

```
(praneetha@Praneetha)~[~/Desktop]
$ grep -E '^[0-9]' security_log.txt
```

- Find lines ending in .exe.

grep -E '\.exe\$' security_log.txt

```
(praneetha@Praneetha)~[~/Desktop]
$ grep -E '\.exe$' security_log.txt
[2025-08-18 08:18:20] Suspicious file download evil.exe from http://malware-test.example.org/evil.exe
[2025-08-18 08:24:25] Suspicious SHA256 hash=275a021bbfb6489e54d471899f7db9d1663fc695ec2fe2a2c4538aabf651fd0f file payload.exe
[2025-08-18 08:25:18] WARNING file payload.exe stored at C:\Users\Public\Downloads\payload.exe
[2025-08-18 08:35:58] Suspicious download http://evil.example.net/dropper.exe to C:\Temp\dropper.exe
[2025-08-18 08:47:04] Suspicious download of file backdoor.exe from https://command.bad-domain.in/backdoor.exe
Suspicious malware file evil.exe
```

- Find lines containing filenames ending in .exe, .sh, or .pdf.

grep -E '[[[:alnum:]]_.-]+\.(exe|sh|pdf)\b' security_log.txt

```
(praneetha@Praneetha)~[~/Desktop]
$ grep -E '[[[:alnum:]]_.-]+\.(exe|sh|pdf)\b' security_log.txt
[2025-08-18 08:18:20] Suspicious file download evil.exe from http://malware-test.example.org/evil.exe
[2025-08-18 08:18:32] ERROR file evil.exe hash MD5=44d88612fea8a8f36de82e1278abb02f
[2025-08-18 08:19:04] INFO file report.pdf saved to /home/analyst/downloads/report.pdf
[2025-08-18 08:19:45] Suspicious file invoice_2025.pdf downloaded from https://phishing-example.com/invoice_2025.pdf
[2025-08-18 08:24:01] Suspicious SHA1 hash=3395856ce81f2b7382dee72602f798b642f14140 file updater.sh
[2025-08-18 08:24:25] Suspicious SHA256 hash=275a021bbfb6489e54d471899f7db9d1663fc695ec2fe2a2c4538aabf651fd0f file payload.exe
[2025-08-18 08:24:46] INFO MD5 hash=e2fc714c4727ee9395f324cd2e7f331f file archive.pdf
```

- Identify lines containing the exact word root, but not words such as rooted.

grep -w "root" security_log.txt

```
(praneetha@Praneetha)~[~/Desktop]
$ grep -w "root" security_log.txt
[2025-08-18 08:12:03] Failed login for user root from 203.0.113.45 port 4444
[2025-08-18 08:12:10] Failed login for user root from 203.0.113.45 port 4444
[2025-08-18 08:12:17] Failed login for user root from 203.0.113.45 port 4444
[2025-08-18 08:27:24] Suspicious crontab created for user root
[2025-08-18 08:42:01] Failed login for user root from 198.51.100.77 port 2222
[2025-08-18 08:42:19] Failed login for user root from 198.51.100.77 port 2222
```

- Display blank lines in the file.

grep '^\$' security_log.txt

```
(praneetha@Praneetha)~[~/Desktop]
$ grep '^$' security_log.txt
```

- Display all nonblank lines.

grep -v '^\$' security_log.txt

```
(praneetha@Praneetha)~[~/Desktop]
$ grep -v '^$' security_log.txt
[2025-08-18 08:10:01] INFO Server startup completed on 192.168.10.5 port 8080
[2025-08-18 08:11:14] INFO User admin logged in from 192.168.10.21
[2025-08-18 08:12:03] Failed login for user root from 203.0.113.45 port 4444
[2025-08-18 08:12:10] Failed login for user root from 203.0.113.45 port 4444
[2025-08-18 08:12:17] Failed login for user root from 203.0.113.45 port 4444
[2025-08-18 08:13:02] WARNING Multiple authentication failures from 203.0.113.45
[2025-08-18 08:14:33] INFO Successful login for user analyst from 10.10.5.23
[2025-08-18 08:15:44] ERROR Connection denied from 198.51.100.77 port 9001
[2025-08-18 08:16:05] Suspicious connection from 198.51.100.77 to evilc2.ddns.net:9001
[2025-08-18 08:16:10] Suspicious outbound request https://evilc2-ddns.net/update.php
```

Part D - Linux Filters and Pipelines

- Extract all IPv4 addresses, sort them, and remove duplicates.

`grep -Eo '([0-9]{1,3}\.){3}[0-9]{1,3}' security_log.txt | sort -V | uniq`

```
(praneetha@Praneetha)~[/Desktop]
$ grep -Eo '([0-9]{1,3}\.){3}[0-9]{1,3}' security_log.txt | sort -V | uniq
1.1.1.1
8.8.8.8
10.1.2.3
10.10.5.23
10.20.30.40
172.16.5.10
172.31.200.9
192.168.1.50
192.168.10.5
```

- Count the number of unique IPv4 addresses present in the log.

`grep -Eo '([0-9]{1,3}\.){3}[0-9]{1,3}' security_log.txt | sort -V | uniq | wc -l`

```
(praneetha@Praneetha)~[/Desktop]
$ grep -Eo '([0-9]{1,3}\.){3}[0-9]{1,3}' security_log.txt | sort -V | uniq | wc -l
13
```

- Extract all IP addresses and display them in sorted order.

`grep -Eo '([0-9]{1,3}\.){3}[0-9]{1,3}' security_log.txt | sort -V`

```
(praneetha@Praneetha)~[/Desktop]
$ grep -Eo '([0-9]{1,3}\.){3}[0-9]{1,3}' security_log.txt | sort -V
1.1.1.1
8.8.8.8
8.8.8.8
10.1.2.3
10.1.2.3
10.1.2.3
10.1.2.3
10.10.5.23
10.20.30.40
172.16.5.10
```

- Determine which IP addresses appear more than once.

`grep -Eo '([0-9]{1,3}\.){3}[0-9]{1,3}' security_log.txt | sort -V | uniq -c | awk '$1 > 1'`

```
(praneetha@Praneetha)~[/Desktop]
$ grep -Eo '([0-9]{1,3}\.){3}[0-9]{1,3}' security_log.txt | sort -V | uniq -c | awk '$1 > 1'
2 8.8.8.8
4 10.1.2.3
3 192.168.10.5
4 192.168.10.21
7 198.51.100.77
6 203.0.113.45
3 203.0.113.99
```

- Determine the IP address that occurs most frequently.

`grep -Eo '([0-9]{1,3}\.){3}[0-9]{1,3}' security_log.txt | sort | uniq -c | sort -nr | head -1`

```
(praneetha@Praneetha)~[/Desktop]
$ grep -Eo '([0-9]{1,3}\.){3}[0-9]{1,3}' security_log.txt | sort | uniq -c | sort -nr | head -1
7 198.51.100.77
```

- Extract all usernames from authentication entries and display unique usernames.

`grep -i "login for user" security_log.txt | grep -Eo 'user [[:alnum:]]+' | awk '{print $2}' | sort -u`

```
(praneetha@Praneetha)~[/Desktop]
$ grep -i "login for user" security_log.txt | grep -Eo 'user [[:alnum:]]+' | awk '{print $2}' | sort -u
admin
analyst
guest
root
```

- Count the number of authentication attempts associated with each username

```
grep -i "login for user" security_log.txt | grep -Eo 'user [[:alnum:]]+' | awk '{print $2}' | sort | uniq -c
```

```
(praneetha@Praneetha)~[/Desktop]
$ grep -i "login for user" security_log.txt | grep -Eo 'user [[:alnum:]]+' | awk '{print $2}' | sort | uniq -c
 3 admin
 1 analyst
 2 guest
 5 root
```

- Extract all HTTP status codes and determine how many times each status code occurs.

```
grep -Eo 'status [0-9]{3}' security_log.txt | awk '{print $2}' | sort | uniq -c
```

```
(praneetha@Praneetha)~[/Desktop]
$ grep -Eo 'status [0-9]{3}' security_log.txt | awk '{print $2}' | sort | uniq -c
 4 200
 1 302
 1 403
 2 404
 2 500
```

- Display the five most frequently occurring IP addresses.

```
grep -Eo '([0-9]{1,3}\.){3}[0-9]{1,3}' security_log.txt | sort | uniq -c | sort -nr | head -5
```

```
(praneetha@Praneetha)~[/Desktop]
$ grep -Eo '([0-9]{1,3}\.){3}[0-9]{1,3}' security_log.txt | sort | uniq -c | sort -nr | head -5
 7 198.51.100.77
 6 203.0.113.45
 4 192.168.10.21
 4 10.1.2.3
 3 203.0.113.99
```

- Display the three most frequently occurring usernames.

```
grep -i "login for user" security_log.txt | grep -Eo 'user [[:alnum:]]+' | awk '{print $2}' | sort | uniq -c | sort -nr | head -3
```

```
(praneetha@Praneetha)~[/Desktop]
$ grep -i "login for user" security_log.txt | grep -Eo 'user [[:alnum:]]+' | awk '{print $2}' | sort | uniq -c | sort -nr | head -3
 5 root
 3 admin
 2 guest
```

- Extract all destination ports and display them in numerical order.

```
grep -Eo 'port [0-9]+' security_log.txt | awk '{print $2}' | sort -n
```

```
(praneetha@Praneetha)~[/Desktop]
$ grep -Eo 'port [0-9]+' security_log.txt | awk '{print $2}' | sort -n
22
22
22
443
2222
2222
2222
2222
2222
3389
4444
```

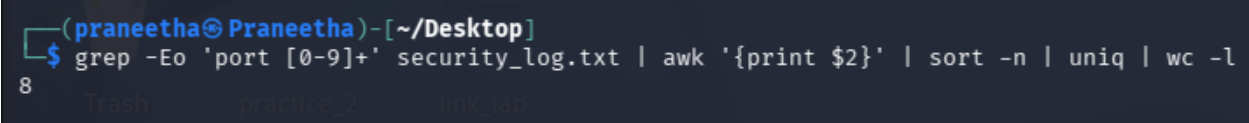
- Remove duplicate port numbers.

```
grep -Eo 'port [0-9]+' security_log.txt | awk '{print $2}' | sort -n | uniq
```

```
(praneetha@Praneetha)~[/Desktop]
$ grep -Eo 'port [0-9]+' security_log.txt | awk '{print $2}' | sort -n | uniq
22
443
2222
3389
4444
8080
8443
9001
```

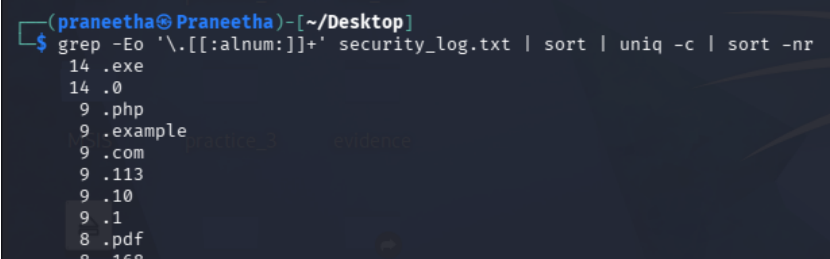
- Count the number of unique destination ports.

```
grep -Eo 'port [0-9]+' security_log.txt | awk '{print $2}' | sort -n | uniq | wc -l
```



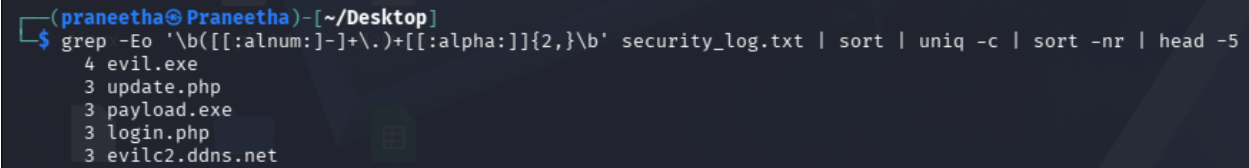
- Extract all file extensions appearing in the log and determine their frequency.

```
grep -Eo '\.[[:alnum:]]+' security_log.txt | sort | uniq -c | sort -nr
```



- Find the five most frequently occurring domains.

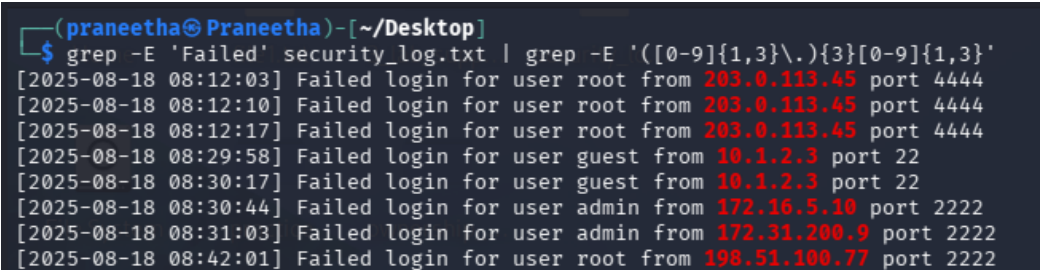
```
grep -Eo '\b([[:alnum:]]+\.[[:alpha:]]{2,})\b' security_log.txt | sort | uniq -c | sort -nr | head -5
```



Part E - Combining grep Conditions

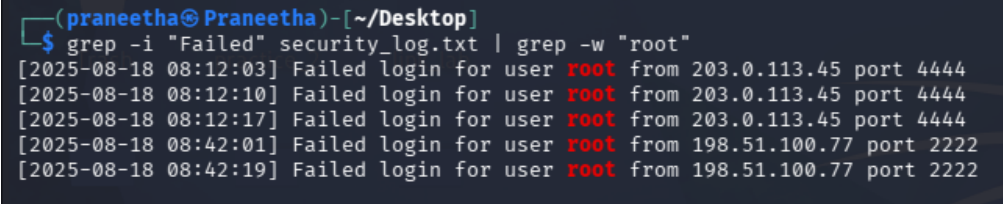
- Display lines containing both an IP address and the word Failed.

```
grep -E 'Failed' security_log.txt | grep -E '([0-9]{1,3}\.){3}[0-9]{1,3}'
```



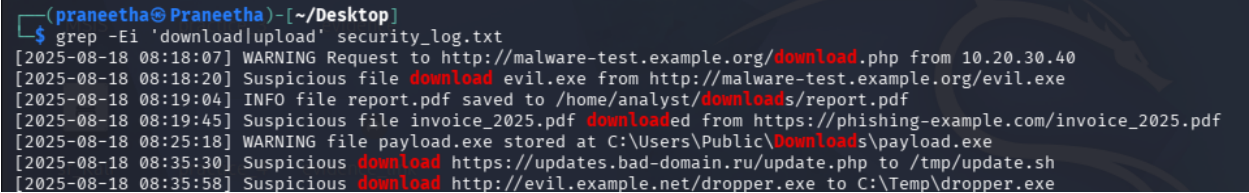
- Display lines containing both root and Failed.

```
grep -i "Failed" security_log.txt | grep -w "root"
```



- Display lines containing either download or upload.

```
grep -Ei 'download|upload' security_log.txt
```



- Display suspicious entries containing either .exe or .sh.

```
grep "Suspicious" security_log.txt | grep -E '\.(exe|sh)\b'
```

```
(praneetha@Praneetha) - [~/Desktop]
$ grep "Suspicious" security_log.txt | grep -E '\.(exe|sh)\b'
[2025-08-18 08:18:20] Suspicious file download evil.exe from http://malware-test.example.org/evil.exe
[2025-08-18 08:24:01] Suspicious SHA1 hash=3395856ce81f2b7382dee72602f798b642f14140 file updater.sh
[2025-08-18 08:24:25] Suspicious SHA256 hash=275a021bbfb6489e54d471899f7db9d1663fc695ec2fe2a2c4538aabf651fd0f file payload.exe
[2025-08-18 08:26:05] Suspicious script stored at /tmp/updater.sh
[2025-08-18 08:35:30] Suspicious download https://updates.bad-domain.ru/update.php to /tmp/update.sh
```

- Display lines containing a CVE identifier and an IP address.

```
grep -E 'CVE-[0-9]{4}-[0-9]{4,}' security_log.txt | grep -E '([0-9]{1,3}\.){3}[0-9]{1,3}'
```

```
(praneetha@Praneetha) - [~/Desktop]
$ grep -E 'CVE-[0-9]{4}-[0-9]{4,}' security_log.txt | grep -E '([0-9]{1,3}\.){3}[0-9]{1,3}'
[2025-08-18 08:22:03] ERROR Exploit attempt CVE-2025-12345 from 203.0.113.99
[2025-08-18 08:47:31] ERROR CVE-2025-54321 exploited from 8.8.8.8 port 8080
```

- Display lines containing both a filename and a hash value.

```
grep -E '\.(exe|sh|pdf)\b' security_log.txt | grep -Ei 'MD5=|SHA1|SHA256'
```

```
(praneetha@Praneetha) - [~/Desktop]
$ grep -E '\.(exe|sh|pdf)\b' security_log.txt | grep -Ei 'MD5=|SHA1|SHA256'
[2025-08-18 08:18:32] ERROR file evil.exe hash MD5=44d88612fea8a8f36de82e1278abb02f
[2025-08-18 08:24:01] Suspicious SHA1 hash=3395856ce81f2b7382dee72602f798b642f14140 file updater.sh
[2025-08-18 08:24:25] Suspicious SHA256 hash=275a021bbfb6489e54d471899f7db9d1663fc695ec2fe2a2c4538aabf651fd0f file payload.exe
[2025-08-18 08:36:19] ERROR file dropper.exe hash SHA256=3a7bd3e2360a3d80a9d0b7b93d7f90d7c0b8c9a3e9f1d2c4b5a6978877665544
[2025-08-18 08:45:49] ERROR file malware.exe hash MD5=5d41402abc4b2a76b9719d911017c592
```

- Display entries containing an IP address followed by a port.

```
grep -E '([0-9]{1,3}\.){3}[0-9]{1,3}:[0-9]+|([0-9]{1,3}\.){3}[0-9]{1,3}.*port [0-9]+' security_log.txt
```

```
(praneetha@Praneetha) - [~/Desktop]
$ grep -E '([0-9]{1,3}\.){3}[0-9]{1,3}:[0-9]+|([0-9]{1,3}\.){3}[0-9]{1,3}.*port [0-9]+' security_log.txt
[2025-08-18 08:10:01] INFO Server startup completed on 192.168.10.5 port 8080
[2025-08-18 08:12:03] Failed login for user root from 203.0.113.45 port 4444
[2025-08-18 08:12:10] Failed login for user root from 203.0.113.45 port 4444
[2025-08-18 08:12:03] Failed login for user root from 203.0.113.45 port 4444
[2025-08-18 08:15:44] ERROR Connection denied from 198.51.100.77 port 9001
[2025-08-18 08:29:11] INFO Connection from 10.1.2.3 port 22
[2025-08-18 08:29:32] INFO Connection from 10.1.2.3 port 22
[2025-08-18 08:29:58] Failed login for user guest from 10.1.2.3 port 22
[2025-08-18 08:30:17] Failed login for user guest from 10.1.2.3 port 22
```

- Display HTTP requests containing both a URL and status code 404.

```
grep -E 'https?://.*status 404' security_log.txt
```

```
(praneetha@Praneetha) - [~/Desktop]
$ grep -E 'https?://.*status 404' security_log.txt
[2025-08-18 08:33:55] WARNING URL https://shop.example.com/admin.php status 404
```

- Display authentication failures but exclude entries generated from 127.0.0.1.

```
grep -i "Failed login" security_log.txt | grep -v "127\.\0\.\0\1"
```

```
(praneetha@Praneetha) - [~/Desktop]
$ grep -i "Failed login" security_log.txt | grep -v "127\.\0\.\0\1"
[2025-08-18 08:12:03] Failed login for user root from 203.0.113.45 port 4444
[2025-08-18 08:12:10] Failed login for user root from 203.0.113.45 port 4444
[2025-08-18 08:12:17] Failed login for user root from 203.0.113.45 port 4444
[2025-08-18 08:29:58] Failed login for user guest from 10.1.2.3 port 22
[2025-08-18 08:30:17] Failed login for user guest from 10.1.2.3 port 22
[2025-08-18 08:30:44] Failed login for user admin from 172.16.5.10 port 2222
```

- Display attack-related entries but exclude informational (INFO) messages.

```
grep -Ei 'attack|exploit|malware|suspicious' security_log.txt | grep -v "INFO"
```

```
(praneetha@Praneetha)~[~/Desktop]
$ grep -Ei 'attack|exploit|malware|suspicious' security_log.txt | grep -v "INFO"
[2025-08-18 08:16:05] Suspicious connection from 198.51.100.77 to evilc2.ddns.net:9001
[2025-08-18 08:16:19] Suspicious outbound request https://evilc2.ddns.net/update.php
[2025-08-18 08:18:07] WARNING Request to http://malware-test.example.org/download.php from 10.20
[2025-08-18 08:18:20] Suspicious file download evil.exe from http://malware-test.example.org/evi
[2025-08-18 08:19:45] Suspicious file invoice_2025.pdf downloaded from https://phishing-example.
[2025-08-18 08:20:10] WARNING Email received from attacker123@phishing-example.com
[2025-08-18 08:20:51] Suspicious email phishing.alert@evilmail.ru contained malicious attachment
[2025-08-18 08:22:03] ERROR Exploit attempt CVE-2025-12345 from 203.0.113.99
```

Part F - Negative Matching

- Extract all IPv4 addresses except those beginning with 127..

```
grep -Eo '([0-9]{1,3}\.){3}[0-9]{1,3}' security_log.txt | grep -v '^127\.'
```

```
(praneetha@Praneetha)~[~/Desktop]
$ grep -Eo '([0-9]{1,3}\.){3}[0-9]{1,3}' security_log.txt | grep -v '^127\.'
192.168.10.5
192.168.10.21
203.0.113.45
203.0.113.45
203.0.113.45
203.0.113.45
10.10.5.23
198.51.100.77
```

- Display all log entries except successful login events.

```
grep -Eiv 'successful login' security_log.txt
```

```
(praneetha@Praneetha)~[~/Desktop]
$ grep -Eiv 'successful login' security_log.txt
[2025-08-18 08:10:01] INFO Server startup completed on 192.168.10.5 port 8080
[2025-08-18 08:11:14] INFO User admin logged in from 192.168.10.21
[2025-08-18 08:12:03] Failed login for user root from 203.0.113.45 port 4444
[2025-08-18 08:12:10] Failed login for user root from 203.0.113.45 port 4444
[2025-08-18 08:12:17] Failed login for user root from 203.0.113.45 port 4444
[2025-08-18 08:13:02] WARNING Multiple authentication failures from 203.0.113.45
[2025-08-18 08:15:44] ERROR Connection denied from 198.51.100.77 port 9001
[2025-08-18 08:16:05] Suspicious connection from 198.51.100.77 to evilc2.ddns.net
```

- Display all entries that do not contain INFO, DEBUG, or SUCCESS.

```
grep -Ev 'INFO|DEBUG|SUCCESS' security_log.txt
```

```
(praneetha@Praneetha)~[~/Desktop]
$ grep -Ev 'INFO|DEBUG|SUCCESS' security_log.txt
[2025-08-18 08:12:03] Failed login for user root from 203.0.113.45 port 4444
[2025-08-18 08:12:10] Failed login for user root from 203.0.113.45 port 4444
[2025-08-18 08:12:17] Failed login for user root from 203.0.113.45 port 4444
[2025-08-18 08:13:02] WARNING Multiple authentication failures from 203.0.113.45
[2025-08-18 08:15:44] ERROR Connection denied from 198.51.100.77 port 9001
[2025-08-18 08:16:05] Suspicious connection from 198.51.100.77 to evilc2.ddns.net:9001
[2025-08-18 08:16:19] Suspicious outbound request https://evilc2.ddns.net/update.php
```

- Find HTTP requests whose status code is not 200.

```
grep -E 'https?:/' security_log.txt | grep -Ev 'status 200'
```

```
(praneetha@Praneetha)~[~/Desktop]
$ grep -E 'https?:/' security_log.txt | grep -Ev 'status 200'
[2025-08-18 08:16:19] Suspicious outbound request https://evilc2.ddns.net/update.php
[2025-08-18 08:18:07] WARNING Request to http://malware-test.example.org/download.php from 10.20.30.40
[2025-08-18 08:18:20] Suspicious file download evil.exe from http://malware-test.example.org/evil.exe
[2025-08-18 08:19:45] Suspicious file invoice_2025.pdf downloaded from https://phishing-example.com/invoice_2025.pdf
[2025-08-18 08:22:48] Suspicious exploit reference CVE-2025-987654 targeting https://target.example.com/login.php
[2025-08-18 08:33:55] WARNING URL https://shop.example.com/admin.php status 404
[2025-08-18 08:34:16] ERROR URL http://legacy.example.net/login.php status 500
```

- **Display suspicious traffic excluding connections to 192.168.x.x.**

```
grep "Suspicious" security_log.txt | grep -Ev '192\.168\.'
```

```
(praneetha@Praneetha) - [~/Desktop]
$ grep "Suspicious" security_log.txt | grep -Ev '192\.168\.'
[2025-08-18 08:16:05] Suspicious connection from 198.51.100.77 to evilc2.ddns.net:9001
[2025-08-18 08:16:19] Suspicious outbound request https://evilc2.ddns.net/update.php
[2025-08-18 08:18:20] Suspicious file download evil.exe from http://malware-test.example.org/evil.e
[2025-08-18 08:19:45] Suspicious file invoice_2025.pdf downloaded from https://phishing-example.com
[2025-08-18 08:20:51] Suspicious email phishing.alert@evilmail.ru contained malicious attachment
[2025-08-18 08:22:48] Suspicious exploit reference CVE-2025-987654 targeting https://target.example
[2025-08-18 08:24:01] Suspicious SHA1 hash=3395856ce81f2b7382dee72602f798b642f14140 file updater.sh
[2025-08-18 08:24:25] Suspicious SHA256 hash=275a021bbfb6489e54d471899f7db9d1663fc695ec2fe2a2c4538a
[2025-08-18 08:26:05] Suspicious script stored at /tmp/updater.sh
[2025-08-18 08:27:02] Suspicious persistence entry added to /etc/cron.d/system_update
```

- **Extract email addresses except those belonging to the organization's domain.**

```
grep -Eo '[[[:alnum:]]._%+~]+@[[[:alnum:]].-]+\.[[:alpha:]]+' security_log.txt | grep -v
 '@example\.com$'
```

```
(praneetha@Praneetha) - [~/Desktop]
$ grep -Eo '[[[:alnum:]]._%+~]+@[[[:alnum:]].-]+\.[[:alpha:]]+' security_log.txt | grep -v '@example\.com$'
attacker123@phishing-example.com
phishing.alert@evilmail.ru
student2025@university.in
```

Part G - Redirection

- **Extract all IPv4 addresses and save them into:**

ips.txt

```
grep -Eo '([0-9]{1,3}\.){3}[0-9]{1,3}' security_log.txt > ips.txt
```

- **Extract all email addresses and save them into:**

emails.txt

```
grep -Eo '[[[:alnum:]]._%+~]+@[[[:alnum:]].-]+\.[[:alpha:]]+' security_log.txt > emails.txt
```

- **Extract all URLs and save them into:**

urls.txt

```
grep -Eo 'https?://[^\[:space:]]+' security_log.txt > urls.txt
```

- **Extract all hashes and save them into:**

hashes.txt

```
grep -Eio '\b[[:xdigit:]]{32}|\b[[:xdigit:]]{40}|\b[[:xdigit:]]{64}' security_log.txt > hashes.txt
```

- **Extract failed-login entries and save them into:**

failed_login.txt

```
grep -i "Failed login" security_log.txt > failed_login.txt
```

- **Extract suspicious entries and append them to:**

incident_report.txt

```
grep "Suspicious" security_log.txt >> incident_report.txt
```

- **Store unique IP addresses in:**

unique_ips.txt

```
grep -Eo '([0-9]{1,3}\.){3}[0-9]{1,3}' security_log.txt | sort -V | uniq > unique_ips.txt
```

- **Store IP-frequency statistics in:**

ip_frequency.txt

```
grep -Eo '([0-9]{1,3}\.){3}[0-9]{1,3}' security_log.txt | sort | uniq -c | sort -nr > ip_frequency.txt
```

- **Redirect normal output to results.txt and errors to errors.txt.**

```
grep -Eo '([0-9]{1,3}\.){3}[0-9]{1,3}' security_log.txt > results.txt 2> errors.txt
```

- **Extract all domains from the log entries containing Domain accessed, convert the domain names to lowercase, sort them alphabetically, remove duplicates, and save the result in domains.txt.**

```
grep "Domain accessed" security_log.txt | grep -Eo '([[:alnum:]]+\.)+[[:alpha:]]{2,}' | tr '[:upper:]' '[:lower:]' | sort -u > domains.txt
```

- **Identify all IP addresses involved in failed-login attempts. Convert the output into a clean list containing one unique IP per line and save it as failed_login_ips.txt.**

```
grep -i "Failed login" security_log.txt | grep -Eo '([0-9]{1,3}\.){3}[0-9]{1,3}' | sort -V -u > failed_login_ips.txt
```

Combining filters

- **Extract all failed-login usernames and save them into:**

failed_users.txt

```
grep -i "Failed login" security_log.txt | grep -Eo 'user [[:alnum:]]+' | awk '{print $2}' > failed_users.txt
```

- **Extract all domains and save them in lowercase into:**

domains.txt

```
grep -Eo '([[:alnum:]]+\.)+[[:alpha:]]{2,}' security_log.txt | tr '[:upper:]' '[:lower:]' | sort -u > domains.txt
```

- Extract all MD5 hashes, convert them to uppercase, and save them into:

md5_hashes.txt

```
grep -Eo '[:xdigit:]{32}' security_log.txt | tr '[:lower:]' '[:upper:]' > md5_hashes.txt
```

- Extract source IP addresses from failed-login records, sort them, remove duplicates, and redirect the result to:

failed_ips.txt

```
grep -i "Failed login" security_log.txt | grep -Eo '([0-9]{1,3}\.){3}[0-9]{1,3}' | sort -V -u > failed_ips.txt
```

- Create a file named clean_log.txt in which all square brackets are removed from the original log.

```
sed 's/\[/g; s/\]/g' security_log.txt > clean_log.txt
```

- Create normalized_log.txt by converting the complete log file to lowercase.

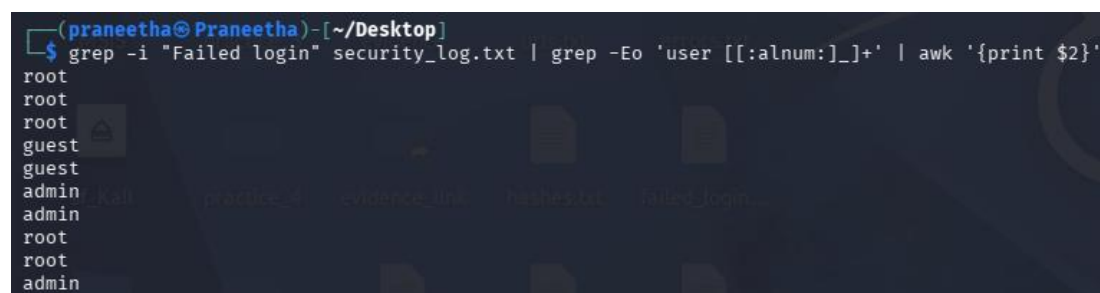
```
tr '[:upper:]' '[:lower:]' < security_log.txt > normalized_log.txt
```

- Extract all domain names, convert them to lowercase, sort them, remove duplicates, and save them to unique_domains.txt.

```
grep -Eo '([[:alnum:]]+\.)+[[:alpha:]]{2,}' security_log.txt | tr '[:upper:]' '[:lower:]' | sort -u > unique_domains.txt
```

- Find all failed-login entries and extract only the usernames.

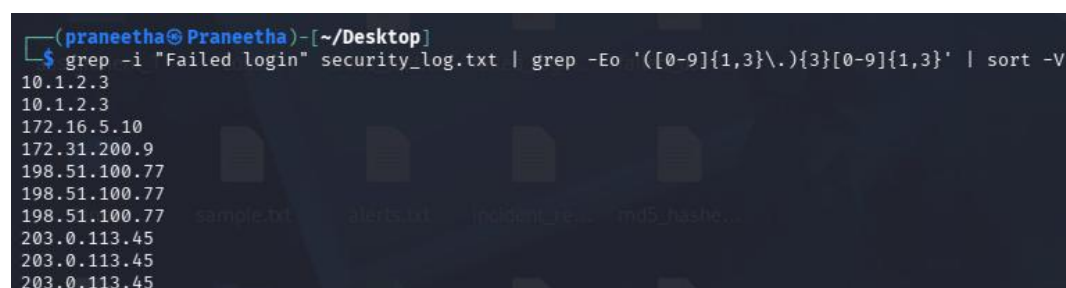
```
grep -i "Failed login" security_log.txt | grep -Eo 'user [[:alnum:]]+' | awk '{print $2}'
```



```
(praneetha@Praneetha) - [~/Desktop]
$ grep -i "Failed login" security_log.txt | grep -Eo 'user [[:alnum:]]+' | awk '{print $2}'
root
root
root
guest
guest
admin
admin
root
root
admin
```

- Find all failed-login entries, extract only the source IP addresses, and display them in sorted order.

```
grep -i "Failed login" security_log.txt | grep -Eo '([0-9]{1,3}\.){3}[0-9]{1,3}' | sort -V
```



```
(praneetha@Praneetha) - [~/Desktop]
$ grep -i "Failed login" security_log.txt | grep -Eo '([0-9]{1,3}\.){3}[0-9]{1,3}' | sort -V
10.1.2.3
10.1.2.3
172.16.5.10
172.31.200.9
198.51.100.77
198.51.100.77
198.51.100.77
203.0.113.45
203.0.113.45
203.0.113.45
```

- Extract IP addresses associated with failed logins and determine the unique IPs.

```
grep -i "Failed login" security_log.txt | grep -Eo '([0-9]{1,3}\.){3}[0-9]{1,3}' | sort -V -u
```

```
(praneetha@Praneetha)~[~/Desktop]
$ grep -i "Failed login" security_log.txt | grep -Eo '([0-9]{1,3}\.){3}[0-9]{1,3}' | sort -V -u
10.1.2.3
172.16.5.10
172.31.200.9
198.51.100.77
203.0.113.45
```

- Extract usernames involved in failed logins, convert them to uppercase, and remove duplicate values.

```
grep -i "Failed login" security_log.txt | grep -Eo 'user [[:alnum:]]+' | awk '{print $2}' | tr '[:lower:]' '[:upper:]' | sort -u
```

```
(praneetha@Praneetha)~[~/Desktop]
$ grep -i "Failed login" security_log.txt | grep -Eo 'user [[:alnum:]]+' | awk '{print $2}' | tr '[:lower:]' '[:upper:]' | sort -u
ADMIN
GUEST
ROOT
```

- Find all lines containing Domain accessed, extract only the domain names, and convert them to lowercase.

```
grep "Domain accessed" security_log.txt | grep -Eo '([[:alnum:]]+\.)+[[:alpha:]]{2,}' | tr '[:upper:]' '[:lower:]'
```

```
(praneetha@Praneetha)~[~/Desktop]
$ grep "Domain accessed" security_log.txt | grep -Eo '([[:alnum:]]+\.)+[[:alpha:]]{2,}' | tr '[:upper:]' '[:lower:]'
example.com
research.org
suspicious-site.ru
command.bad-domain.in
portal.university.in
```

- Find all lines containing MD5=, extract only the MD5 hash values, and convert hexadecimal letters to uppercase.

```
grep "MD5=" security_log.txt | grep -Eo 'MD5=[[:xdigit:]]{32}' | cut -d= -f2 | tr '[:lower:]' '[:upper:]'
```

```
(praneetha@Praneetha)~[~/Desktop]
$ grep "MD5=" security_log.txt | grep -Eo 'MD5=[[:xdigit:]]{32}' | cut -d= -f2 | tr '[:lower:]' '[:upper:]'
44D88612FEA8A8F36DE82E1278ABB02F
5D41402ABC4B2A76B9719D911017C592
```

- Find entries containing User-Agent:, extract the user-agent field, and convert it to lowercase.

```
grep "User-Agent:" security_log.txt | sed 's/.*User-Agent: //' | tr '[:upper:]' '[:lower:]'
```

```
(praneetha@Praneetha)~[~/Desktop]
$ grep "User-Agent:" security_log.txt | sed 's/.*User-Agent: //' | tr '[:upper:]' '[:lower:]'
mozilla/5.0 evilsscanner/2.1
mozilla/5.0 chrome/139.0
curl/8.5.0
python-requests/2.31.0
```

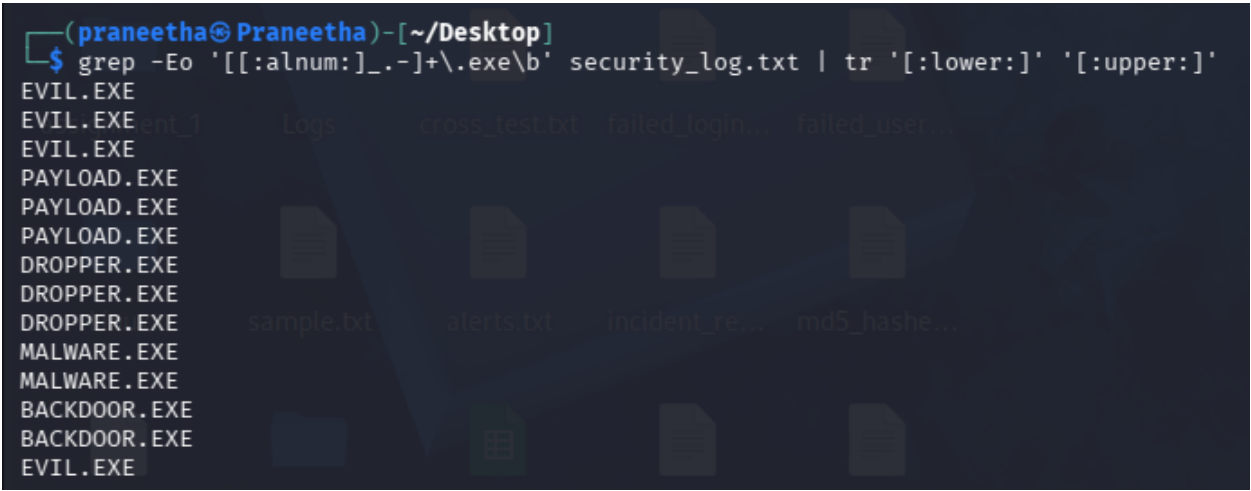
- Find all CVE-related lines, extract the CVE identifiers, and convert them to lowercase.

```
grep -Eo 'CVE-[0-9]{4}-[0-9]{4}' security_log.txt | tr '[:upper:]' '[:lower:]'
```

```
(praneetha@Praneetha)~[~/Desktop]
$ grep -Eo 'CVE-[0-9]{4}-[0-9]{4}' security_log.txt | tr '[:upper:]' '[:lower:]'
cve-2025-12345
cve-2024-3094
cve-2025-987654
cve-2023-44487
cve-2025-54321
cve-2025-54321
```

- Find all entries containing .exe, extract the relevant filename, and convert it to uppercase.

```
grep -Eo '[[[:alnum:]]_.-]+\.\exe\b' security_log.txt | tr '[:lower:]' '[:upper:]'
```



- Extract all domains from Domain accessed entries, convert them to uppercase, sort them, and remove duplicates.

```
grep "Domain accessed" security_log.txt | grep -Eo '([[[:alnum:]]-]+\.\.)([[:alpha:]]{2,})' | tr '[:lower:]' '[:upper:]' | sort -u
```

